

Herpes Zoster: Clinical Features and Complications

Dermatomal Involvement and Rash Evolution

- **Early stage:** Characterized by erythema within a thoracic dermatome, followed by grouped vesicle formation.
- **Later stage:** Crusted lesions appear on the back (site of initial eruption), with newer, confluent hemorrhagic vesicles and bullae on the lateral chest wall. Vesicles extending beyond the primary dermatome suggest hematogenous dissemination, which is not uncommon.
- Crusting typically persists for 2–3 weeks.
- New lesions usually appear for 1–4 days, occasionally up to 7 days in immunocompetent individuals.
- Rash severity and duration are greatest in older adults and mildest in children.

Ophthalmic Zoster

- Involves the ophthalmic division of the trigeminal nerve in 10%–15% of herpes zoster cases.
- Rash distribution extends from the eye to the skull vertex, stopping sharply at the midline of the forehead.
- Isolated involvement of the **supraorbital** and **supratrochlear** nerves usually spares the eye.
- **Nasociliary nerve** involvement (which innervates the eye and nasal tip) increases risk of ocular complications.
- **Hutchinson's sign:** Vesicles on the tip or side of the nose indicate potential ocular involvement, necessitating ophthalmologic evaluation.
- Ocular involvement occurs in 30%–40% of ophthalmic zoster cases.
- Common complications include impaired corneal sensation, **neurotrophic keratitis**, and chronic corneal ulceration.

Cranial Nerve Involvement

- Herpes zoster affecting the **mandibular** and **maxillary** divisions of the trigeminal nerve or other cranial nerves may cause lesions in the mouth, pharynx, larynx, or ears.
- **Ramsay Hunt syndrome:** Results from facial (VII) and auditory (VIII) nerve involvement.
- Presents with facial palsy, vesicles on the external ear or tympanic membrane.
- May include tinnitus, vertigo, and sensorineural hearing loss.

Pain in Herpes Zoster

- Pain is the most significant symptom, particularly in elderly patients.
- Acute-phase pain (within 30 days of rash onset) ranges from mild to severe.
- Described as burning, aching, tingling, itching, or stabbing.
- Some patients report excruciating or unbearable pain.
- Associated with reduced physical function, emotional distress, and impaired social functioning.

Herpes Zoster in Immunocompromised Hosts

- Immunocompromised individuals are at higher risk for severe complications beyond postherpetic neuralgia (PHN).
- Common complications:
 - Skin necrosis and scarring.
 - Cutaneous dissemination (25%–50% incidence).
 - Visceral dissemination (in ~10% of disseminated cases), often involving lungs, liver, and brain; may be fatal.
- **HIV-infected patients:**
 - Prone to multiple recurrences of herpes zoster as immune function declines.
 - Recurrences may occur in the same, different, or multiple contiguous/non-contiguous dermatomes.

- In *AIDS*, zoster can be severe, with widespread cutaneous and visceral dissemination.